Diabetes Prediction with Machine Learning

ICT302 Advanced Data Analytics for Business – Assessment 2 · Guiding Questions

Dataset: CDC BRFSS 2015 Diabetes Health Indicators (archive (2)) · Notebook: Diabetes_ML_Model.ipynb

1. Problem Framing

What is the business problem being addressed?

Health insurers, GP networks and public-health agencies need a low-cost way to identify people who are likely to have diabetes but have not been diagnosed, so they can be invited for a blood test (HbA1c). The question is: **can routine health-survey answers predict whether a person has diabetes?**

Why is it important to solve?

- **Common and under-diagnosed:** 13.9% of BRFSS 2015 respondents report diabetes and a further 1.8% pre-diabetes; many more cases are undiagnosed.
- **Costly:** late diagnosis leads to expensive complications (heart disease, kidney failure, amputation, blindness). Early lifestyle or drug treatment is far cheaper.
- **Resource allocation:** testing everyone is expensive. A risk score lets an organisation focus limited testing and outreach budgets on the highest-risk people.

What type of machine learning is most suitable?

Supervised learning – classification. Every record carries a known label, so models learn the mapping from features to outcome and predict it for new people. The main task is **binary** classification (diabetes yes/no); an extension uses **multi-class** classification (none / pre-diabetes / diabetes). Unsupervised learning (e.g. clustering) could segment patients but would not predict the outcome directly, and reinforcement learning does not fit because there is no sequential action-reward loop.

2. Data Selection & Preparation

Which dataset is chosen and why?

The **CDC BRFSS 2015 Diabetes Health Indicators** dataset: answers to the US Behavioral Risk Factor Surveillance System telephone survey, cleaned to 21 health-indicator features. It is supplied in three files:

File	Rows	Target	Use in this study
diabetes_binary_health_indicators_BRFSS2015	253,680	Diabetes_binary: 0 = no diabetes or pre-diabetes, 1 = diabetes (13.9%)	Main model (Part A)
diabetes_binary_5050split_health_indicators_BRFSS2015	70,692	Same label, balanced 50/50 (all 35,346 diabetics + equal random sample of non-diabetics)	Does balancing help? (Part B)
diabetes_012_health_indicators_BRFSS2015	253,680	Diabetes_012: 0 = none (84.2%), 1 = pre-diabetes (1.8%), 2 = diabetes (13.9%)	Can pre-diabetes be separated? (Part C)

Why: it is large and real-world, has no missing values, and its features are cheap to collect, which matches a screening use case. The three versions allow two ways of handling class imbalance and two problem framings to be compared.

What are the key variables/features?

Group	Features
Health status	GenHlth (self-rated 1–5), MentHlth, PhysHlth (poor-health days in last 30), DiffWalk
Clinical risk flags	HighBP, HighChol, CholCheck, BMI, Stroke, HeartDiseaseorAttack
Lifestyle	PhysActivity, Smoker, HvyAlcoholConsump, Fruits, Veggies
Demographic / socio-economic	Age (13 five-year bands), Sex, Education (1–6), Income (1–8), AnyHealthcare, NoDocbcCost

Strongest correlations with diabetes: GenHlth (0.29), HighBP (0.26), DiffWalk (0.22), BMI (0.22), HighChol (0.20), Age (0.18). Income (-0.16) and physical activity (-0.12) are protective. No single feature exceeds 0.3, so a multivariate model is required.

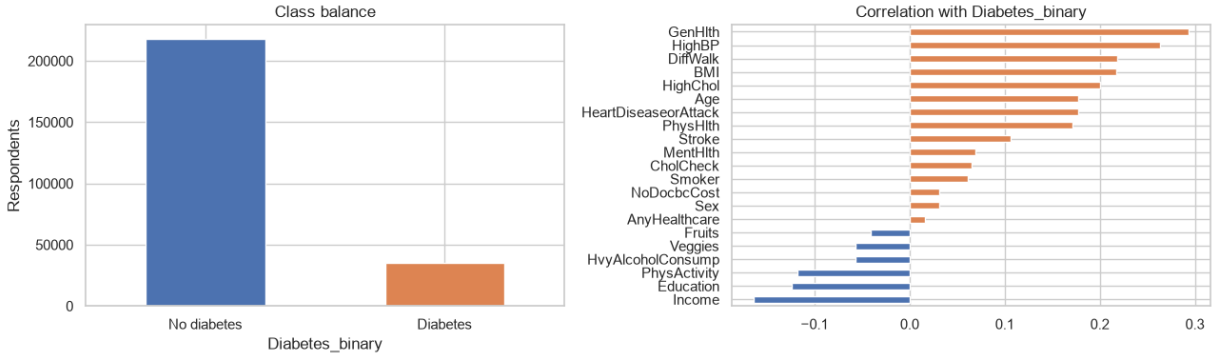


Figure 1 – Class balance and correlation of each feature with Diabetes_binary.

What preprocessing steps are required?

Step	What was done and why
Missing values	None present – no imputation required.
Duplicates	~24,000 identical rows kept: with mostly yes/no answers, different respondents can give identical answers, so they are genuine observations.
Data types	Float-coded values converted to integers.
Encoding	Not required – features are already binary flags or ordinal scales (e.g. Age 1–13).
Scaling	StandardScaler for Logistic Regression only; tree models are scale-invariant.
Class imbalance	Two approaches compared: class_weight="balanced" on the full data, and training on the 50/50 balanced file.
Train/test split	Stratified 80/20 split (202,944 / 50,736 rows), keeping the 13.9% positive rate in both.
Leakage control	Scaling and threshold selection learned on training data only, inside sklearn Pipelines and cross-validation. The 50/50 file is a subset of the same rows, so it is evaluated with its own cross-validation rather than the main test set.

3. Modeling Approaches

Which algorithms are applied, and what are their advantages and disadvantages?

Algorithm	Advantages in this context	Disadvantages in this context
Dummy (majority class)	Baseline that shows what "no skill" looks like	Never predicts diabetes
Logistic Regression	Fast (0.3 s to train); coefficients become odds ratios that clinicians and managers understand; well-calibrated probabilities	Assumes linear log-odds; misses interactions (e.g. age × BMI) unless engineered
Decision Tree (depth 8)	Highly interpretable if-then rules; captures non-linearity	Unstable – small data changes alter the tree; weakest model (AUC 0.817)
Random Forest (200 trees)	Robust; captures interactions; less overfitting than a single tree	Slower (5.4 s to train); black box; large model file
Histogram Gradient Boosting (tuned)	Highest accuracy; fast on large data (2 s to train); handles non-linear effects	Least interpretable; needs hyper-parameter tuning; can overfit without regularisation

Gradient Boosting was tuned with a 20-iteration randomised search (3-fold CV). Best settings: learning_rate 0.05, 200 iterations, 15 leaf nodes, min_samples_leaf 200. The tuned model was also applied to the 50/50 file and to the 3-class problem. SVMs were not used because they scale poorly to ~200k rows; neural networks need more tuning, are harder to interpret, and rarely beat gradient boosting on tabular yes/no-style data.

4. Evaluation

Which performance metrics are used?

- **Accuracy is misleading here:** the dummy model scores **86% accuracy with 0% recall**, so accuracy is reported but not used for model selection.
- **ROC-AUC** (main metric) – how well the model ranks diabetics above non-diabetics, independent of the decision threshold.
- **PR-AUC (area under the precision-recall curve)** – suited to imbalanced data (no-skill baseline = 0.14).
- **Recall** – share of real diabetics caught (most important for screening).
- **Precision** – share of flagged people who really have diabetes (wasted referrals).
- **F1** – balances precision and recall; **macro F1 / macro ROC-AUC** for the 3-class model.
- RMSE is not applicable because it is a regression metric.

How do the models compare?

Part A – 5-fold cross-validation on the training set (full binary file)

Model	ROC-AUC	PR-AUC	Recall	Precision	F1
Gradient Boosting	0.830	0.435	0.794	0.307	0.443
Random Forest	0.829	0.432	0.779	0.311	0.445
Logistic Regression	0.823	0.406	0.766	0.312	0.443
Decision Tree	0.817	0.403	0.781	0.297	0.430
Dummy	0.500	0.139	0.000	0.000	0.000

Part A – tuned Gradient Boosting on the held-out test set (ROC-AUC 0.827, PR-AUC 0.423)

Threshold	Recall	Precision	Accuracy	F1
0.50 (default)	79.5%	30.6%	72.0%	0.442
0.66 (maximises F1)	60.2%	38.1%	80.8%	0.466
0.49 (recall at least 80%)	80.3%	30.2%	71.4%	0.439

The test ROC-AUC (0.827) is almost identical to the CV score (0.830), so the model is not overfitting.

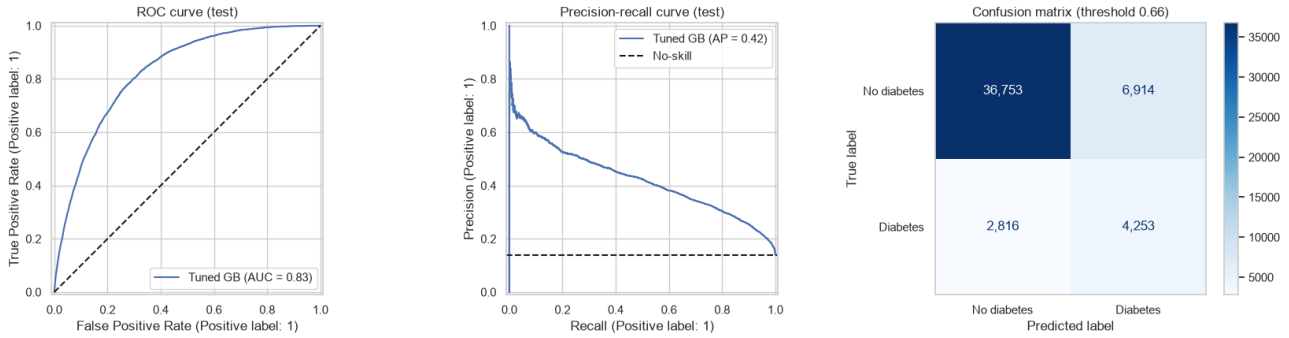


Figure 2 – ROC curve, precision-recall curve and confusion matrix (threshold 0.66) on the test set.

Part B – balanced 50/50 file (5-fold cross-validation)

Model	ROC-AUC	Recall	Precision	Accuracy
Gradient Boosting	0.831	0.799	0.732	0.753
Random Forest	0.828	0.794	0.731	0.750
Logistic Regression	0.825	0.767	0.739	0.748

ROC-AUC is unchanged versus the full data, so balancing does not improve the model. Precision and accuracy look much higher only because half the file is diabetic; at the real ~14% prevalence precision would fall back to about 30–38%.

Part C – 3-class model on the test set (macro ROC-AUC 0.776, macro F1 0.42)

Class	Precision	Recall	Test cases
None	0.96	0.63	42,735
Pre-diabetes	0.03	0.35	932
Diabetes	0.35	0.62	7,069

Pre-diabetes cannot be identified reliably from survey answers: only 3% of pre-diabetes predictions are correct.

What trade-offs exist?

- **Interpretability vs. accuracy:** the top four models are only 0.013 AUC apart. Logistic Regression gives up very little and is far easier to explain – important in healthcare – so it is a strong deployment candidate even though Gradient Boosting scores highest.
- **Recall vs. precision (threshold):** moving from 0.66 to 0.49 catches about 20 percentage points more diabetics but lowers precision from 38% to 30%, meaning more unnecessary tests. The choice depends on the cost of a test versus the cost of a missed case.
- **Computational cost:** Logistic Regression 0.3 s, Gradient Boosting 2 s, Random Forest 5.4 s per training run. All are cheap at this scale; Random Forest has the largest model and slowest predictions.

- **Data balancing vs. honest metrics:** training on balanced data inflates accuracy and precision without improving the model; results must be reported at real-world prevalence.
- **Fairness:** Sex, Age, Income and Education are inputs, so the model could under-serve or over-flag some groups. Per-group performance was not checked in this study and should be before deployment.

5. Discussion & Conclusion

What are the business insights?

- **A short questionnaire can screen for diabetes** with ROC-AUC of about 0.83, without a blood test, and could be delivered cheaply through insurer apps, pharmacies or online health checks.
- **Main risk drivers** (permutation importance): poor self-rated general health, high BMI, older age, high blood pressure and high cholesterol.
- **Socio-economic gradient:** diabetes prevalence falls steadily as income and education rise, so outreach should target lower-income communities. Physical activity is protective, supporting prevention programmes.
- **The threshold is a business decision, not a technical one:** it can be tuned to the testing budget and the tolerance for missed cases.
- **Balancing the data does not help:** class weighting on the full data ranks people just as well and keeps all 218k non-diabetic records.
- **Pre-diabetes needs a blood test:** survey answers cannot single it out (3% precision), so the realistic design is two steps – survey screening first, then an HbA1c test to separate pre-diabetes from diabetes.

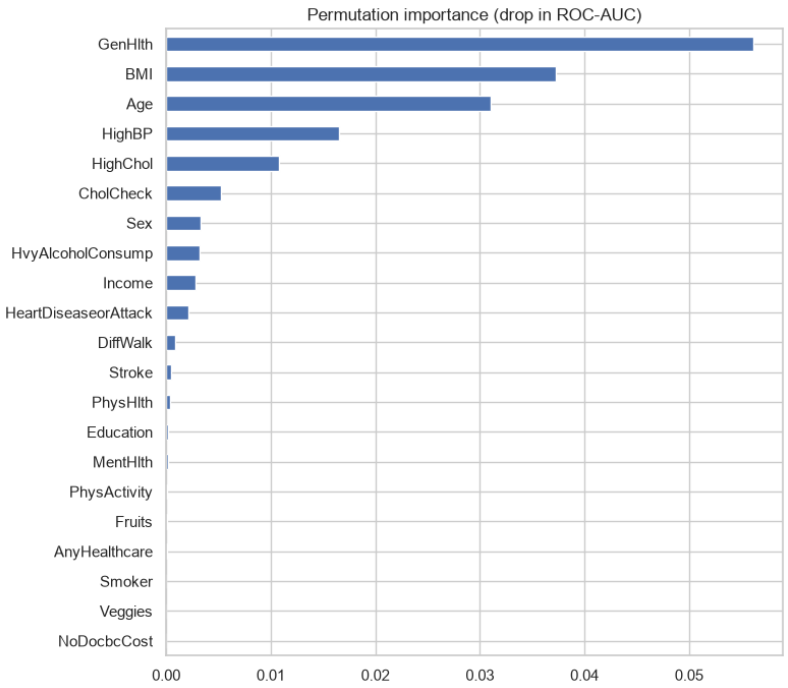


Figure 3 – Permutation importance: drop in test ROC-AUC when each feature is shuffled.

What are the limitations of the study?

- **Self-reported labels:** recall bias, and undiagnosed diabetics are recorded as 0, which adds label noise.
- **Population and time:** 2015 US adults only; results may not transfer to other countries or years.

- **Coarse features:** mostly yes/no answers, with age in 5-year bands and income in broad ranges.
- **Correlation, not causation:** the model cannot say that lowering BMI causes lower risk.
- **Modest precision (30–38%):** most flagged people will not have diabetes – a screening tool, not a diagnosis.
- **Few pre-diabetes examples** (~4.6k) limit the 3-class model.
- **Not assessed:** fairness across groups, and calibration of predicted probabilities.

What further improvements or future work can be done?

- Improve pre-diabetes detection: resampling (e.g. SMOTE), a dedicated pre-diabetes vs. diabetes model for flagged people, or adding lab data.
- Check fairness and performance by sex, age and income group.
- Calibrate probabilities so risk scores are meaningful to patients.
- Try XGBoost / LightGBM and use SHAP values to explain individual predictions.
- Set a cost-based threshold using real testing costs and treatment savings.
- Validate on newer BRFSS years (e.g. 2021–2023) or other countries' data.
- Deploy as a simple web risk calculator.